Taller — Consultas Básicas en PostgreSQL

Universidad de Pamplona — Facultad de Ingenierías y Arquitectura
Programa de Ingeniería de Sistemas
Bases de Datos I — 2026-2
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Cuando un resultado tiene más de 10 filas se muestran las primeras 10 y el total que reporta el cliente. El resultado completo está en el archivo `.sql` de cada literal.

Ejercicio 1. SELECT y alias — Catálogo de departamentos

Tabla: `geo.departments`.

Literal a)

Todas las columnas y filas de la tabla con SELECT *. Archivo: `ex_01_a.sql`.

```
SELECT *
FROM geo.departments AS T1;
```

Resultado:

```
 department_code |                    department_name
-----------------+------------------------------------------------------------
 05              | Antioquia
 08              | Atlántico
 11              | Bogotá, D.C.
 13              | Bolívar
 15              | Boyacá
 17              | Caldas
 18              | Caquetá
 19              | Cauca
 20              | Cesar
 23              | Córdoba
 ...
(33 rows)
```

Literal b)

Código y nombre del departamento, con los alias code y department. Archivo: `ex_01_b.sql`.

```
SELECT
    T1.department_code AS code,
    T1.department_name AS department
FROM
    geo.departments AS T1;
```

Resultado:

```
 code |                       department
------+------------------------------------------------------------
 05   | Antioquia
 08   | Atlántico
 11   | Bogotá, D.C.
 13   | Bolívar
 15   | Boyacá
 17   | Caldas
 18   | Caquetá
 19   | Cauca
 20   | Cesar
 23   | Córdoba
 ...
(33 rows)
```

Literal c)

El literal b) ordenado alfabéticamente por nombre del departamento. Archivo: `ex_01_c.sql`.

```
SELECT
    T1.department_code AS code,
    T1.department_name AS department
FROM
    geo.departments AS T1
ORDER BY T1.department_name ASC;
```

Resultado:

```
 code |                        department
------+-----------------------------------------------------------
 91   | Amazonas
 05   | Antioquia
 81   | Arauca
 88   | Archipiélago De San Andrés, Providencia Y Santa Catalina
 08   | Atlántico
 11   | Bogotá, D.C.
 13   | Bolívar
 15   | Boyacá
 17   | Caldas
 18   | Caquetá
 ...
(33 rows)
```

Ejercicio 2. WHERE con booleanos — Variedades de café

Tabla: `farm.coffee_varieties`.

Literal a)

Nombre, especie y descripción de las variedades resistentes a la roya. Archivo: `ex_02_a.sql`.

```
SELECT
    T1.variety_name AS variety,
    T1.species AS species,
    T1.description AS description
FROM
    farm.coffee_varieties AS T1
WHERE
    T1.is_rust_resistant = TRUE;
```

Resultado:

```
  variety   | species |                            description
------------+---------+-------------------------------------------------------------------
 Castillo   | arabica | Variedad colombiana resistente a la roya, la más sembrada del país.
 Colombia   | arabica | Variedad compuesta resistente a la roya, antecesora de Castillo.
 Cenicafé 1 | arabica | Variedad de porte bajo con resistencia a la roya.
 Tabi       | arabica | Variedad de porte alto resistente a la roya.
 Robusta    | robusta | Especie Coffea canephora, de mayor contenido de cafeína.
(5 rows)
```

Literal b)

Variedades no resistentes a la roya de la especie arabica, por nombre descendente. Archivo: `ex_02_b.sql`.

```
SELECT
    T1.variety_name AS variety,
    T1.species AS species,
    T1.description AS description
FROM
    farm.coffee_varieties AS T1
WHERE
    T1.is_rust_resistant = FALSE
  AND
    T1.species = 'arabica'
ORDER BY
    T1.variety_name DESC;
```

Resultado:

```
    variety    | species |                      description
---------------+---------+-----------------------------------------------------------
 Típica        | arabica | Variedad tradicional de porte alto.
 Pacamara      | arabica | Cruce de Pacas y Maragogipe, grano grande.
 Maragogipe    | arabica | Variedad de grano grande.
 Geisha        | arabica | Variedad de especialidad con perfil floral.
 Caturra       | arabica | Mutación de Borbón de porte bajo y alta productividad.
 Borbón Rosado | arabica | Variedad de especialidad de fruto rosado.
 Borbón        | arabica | Variedad tradicional de buena calidad en taza.
(7 rows)
```

Ejercicio 3. Columnas calculadas — Primas de certificación

Tabla: `farm.certifications`.

Literal a)

Certificación, organismo y prima por kg, de la prima más alta a la más baja. Archivo: `ex_03_a.sql`.

```
SELECT
    T1.certification_name AS certification,
    T1.issuing_body AS issuer,
    T1.premium_per_kg AS premium_per_kg_cop
FROM
    farm.certifications AS T1
ORDER BY
    T1.premium_per_kg DESC;
```

Resultado:

```
    certification    |              issuer               | premium_per_kg_cop
---------------------+-----------------------------------+--------------------
 Orgánico            | Ente certificador orgánico        |             800.00
 Bird Friendly       | Smithsonian Migratory Bird Center |             500.00
 Fairtrade           | Fairtrade International           |             400.00
 Nespresso AAA       | Nespresso / Rainforest Alliance   |             350.00
 Rainforest Alliance | Rainforest Alliance               |             300.00
 C.A.F.E. Practices  | Starbucks / SCS Global Services   |             250.00
 4C                  | 4C Services                       |             150.00
(7 rows)
```

Literal b)

Se agrega la prima por carga (premium_per_kg * 125). Archivo: `ex_03_b.sql`.

```
SELECT
    T1.certification_name AS certification,
    T1.issuing_body AS issuer,
    T1.premium_per_kg AS premium_per_kg_cop,
    (T1.premium_per_kg * 125) AS premium_per_load_cop
FROM
    farm.certifications AS T1
ORDER BY
    T1.premium_per_kg DESC;
```

Resultado:

```
    certification    |              issuer               | premium_per_kg_cop | premium_per_load_cop
---------------------+-----------------------------------+--------------------+----------------------
 Orgánico            | Ente certificador orgánico        |             800.00 |            100000.00
 Bird Friendly       | Smithsonian Migratory Bird Center |             500.00 |             62500.00
 Fairtrade           | Fairtrade International           |             400.00 |             50000.00
 Nespresso AAA       | Nespresso / Rainforest Alliance   |             350.00 |             43750.00
 Rainforest Alliance | Rainforest Alliance               |             300.00 |             37500.00
 C.A.F.E. Practices  | Starbucks / SCS Global Services   |             250.00 |             31250.00
 4C                  | 4C Services                       |             150.00 |             18750.00
(7 rows)
```

Literal c)

Solo certificaciones con prima por kg mayor o igual a 350 COP. Archivo: `ex_03_c.sql`.

```
SELECT
    T1.certification_name AS certification,
    T1.issuing_body AS issuer,
    T1.premium_per_kg AS premium_per_kg_cop,
    (T1.premium_per_kg * 125) AS premium_per_load_cop
FROM
    farm.certifications AS T1
WHERE
    T1.premium_per_kg >= 350
ORDER BY
    T1.premium_per_kg DESC;
```

Resultado:

```
 certification |              issuer               | premium_per_kg_cop | premium_per_load_cop
---------------+-----------------------------------+--------------------+----------------------
 Orgánico      | Ente certificador orgánico        |             800.00 |            100000.00
 Bird Friendly | Smithsonian Migratory Bird Center |             500.00 |             62500.00
 Fairtrade     | Fairtrade International           |             400.00 |             50000.00
 Nespresso AAA | Nespresso / Rainforest Alliance   |             350.00 |             43750.00
(4 rows)
```

Ejercicio 4. WHERE con LIKE — Municipios de Norte de Santander

Tabla: `geo.municipalities`.

Literal a)

Código y nombre de los municipios de Norte de Santander, en orden alfabético. Archivo: `ex_04_a.sql`.

```
SELECT
    T1.municipality_code AS municipality_code,
    T1.municipality_name AS municipality
FROM
    geo.municipalities AS T1
WHERE
    T1.department_code LIKE '54'
ORDER BY
    T1.municipality_name ASC;
```

Resultado:

```
 municipality_code |   municipality
-------------------+-------------------
 54003             | Ábrego
 54051             | Arboledas
 54099             | Bochalema
 54109             | Bucarasica
 54128             | Cáchira
 54125             | Cácota
 54172             | Chinácota
 54174             | Chitagá
 54206             | Convención
 54001             | Cúcuta
 ...
(40 rows)
```

Literal b)

Municipios del mismo departamento cuyo nombre comienza con San. Archivo: `ex_04_b.sql`.

```
SELECT
    T1.municipality_code AS municipality_code,
    T1.municipality_name AS municipality
FROM
    geo.municipalities AS T1
WHERE
    T1.department_code LIKE '54'
AND
    T1.municipality_name LIKE 'San%'
ORDER BY
T1.municipality_name ASC;
```

Resultado:

```
 municipality_code | municipality
-------------------+--------------
 54670             | San Calixto
 54673             | San Cayetano
 54680             | Santiago
(3 rows)
```

Literal c)

Municipios de todo el país cuyo nombre contiene Pamplona, con ILIKE. Archivo: `ex_04_c.sql`.

```
SELECT
    T1.municipality_code AS municipality_code,
    T1.municipality_name AS municipality
FROM
    geo.municipalities AS T1
WHERE
    T1.municipality_name ILIKE '%Pamplona%'
ORDER BY
    T1.municipality_name ASC;
```

Resultado:

```
 municipality_code | municipality
-------------------+--------------
 54518             | Pamplona
(1 row)
```

Ejercicio 5. IS NULL y operadores lógicos — Puntos de compra

Tabla: `buy.collection_points`.

Literal a)

Nombre, dirección y municipio de los puntos de compra inactivos. Archivo: `ex_05_a.sql`.

```
SELECT
    T1.point_name AS point,
    T1.address_line AS address,
    T1.municipality_code AS municipality_code
FROM
    buy.collection_points AS T1
WHERE
    T1.is_active = FALSE;
```

Resultado:

```
           point            |      address       | municipality_code
----------------------------+--------------------+-------------------
 Punto de Compra El Banco 1 | Calle 17 # 34-54   | 47245
 Punto de Compra Popayán 1  | Carrera 21 # 4-95  | 19001
 Punto de Compra Apía 2     | Calle 9 # 14-56    | 66045
 Punto de Compra Tello 1    | Calle 8 # 13-23    | 41799
 Punto de Compra Turbo 1    | Avenida 32 # 34-61 | 05837
 Punto de Compra Paicol 2   | Calle 2 # 30-84    | 41518
 Punto de Compra Rionegro 1 | Carrera 31 # 59-80 | 68615
 Punto de Compra Ulloa 1    | Calle 15 # 3-50    | 76845
 Punto de Compra Nátaga 1   | Diagonal 30 # 7-59 | 41483
(9 rows)
```

Literal b)

Puntos de compra sin teléfono registrado, ordenados por nombre. Archivo: `ex_05_b.sql`.

```
SELECT
    T1.point_name AS point,
    T1.phone_number AS phone_number
FROM
    buy.collection_points AS T1
WHERE
    T1.phone_number IS NULL
ORDER BY
    T1.point_name ASC;
```

Resultado:

```
             point              | phone_number
--------------------------------+--------------
 Punto de Compra Aipe 1         |
 Punto de Compra Anserma 1      |
 Punto de Compra Apía 2         |
 Punto de Compra Baraya 1       |
 Punto de Compra Bochalema 1    |
 Punto de Compra Bosconia 1     |
 Punto de Compra Cajibío 1      |
 Punto de Compra Cisneros 1     |
 Punto de Compra Coello 1       |
 Punto de Compra El Carmen 2    |
 ...
(26 rows)
```

Literal c)

Puntos inactivos o sin teléfono, mostrando phone_number e is_active. Archivo: `ex_05_c.sql`.

```
SELECT
    T1.point_name AS point,
    T1.address_line AS address,
    T1.municipality_code AS municipality_code,
    T1.phone_number AS phone_number,
    T1.is_active AS is_active
FROM
    buy.collection_points AS T1
WHERE
    T1.is_active = FALSE
   OR
    T1.phone_number IS NULL
ORDER BY
    T1.point_name ASC;
```

Resultado:

```
             point              |        address         | municipality_code | phone_number | is_active
--------------------------------+------------------------+-------------------+--------------+-----------
 Punto de Compra Aipe 1         | Carrera 13 # 17-16     | 41016             |              | t
 Punto de Compra Anserma 1      | Calle 39 # 7-96        | 17042             |              | t
 Punto de Compra Apía 2         | Calle 9 # 14-56        | 66045             |              | f
 Punto de Compra Baraya 1       | Calle 1 # 19-77        | 41078             |              | t
 Punto de Compra Bochalema 1    | Carrera 25 # 47-89     | 54099             |              | t
 Punto de Compra Bosconia 1     | Carrera 26 # 26-43     | 20060             |              | t
 Punto de Compra Cajibío 1      | Carrera 15 # 32-45     | 19130             |              | t
 Punto de Compra Cisneros 1     | Calle 37 # 10-5        | 05190             |              | t
 Punto de Compra Coello 1       | Transversal 20 # 49-66 | 73200             |              | t
 Punto de Compra El Banco 1     | Calle 17 # 34-54       | 47245             | 6055026534   | f
 ...
(34 rows)
```

Ejercicio 6. Filtros combinados — Asociados de Pamplona

Tabla: `farm.farmers`.

Literal a)

Asociadas del municipio 54518, ordenadas por apellido y luego por nombre. Archivo: `ex_06_a.sql`.

```
SELECT
    T1.document_number AS document,
    T1.first_name AS first_name,
    T1.last_name AS last_name,
    T1.phone_number AS phone
FROM
    farm.farmers AS T1
WHERE
    T1.gender = 'F'
  AND
    T1.municipality_code = '54518'
ORDER BY
    T1.last_name ASC, T1.first_name ASC;
```

Resultado:

```
  document  |    first_name     |     last_name     |   phone
------------+-------------------+-------------------+------------
 768461420  | Marleny           | Agudelo Sandoval  | 3242280898
 931079279  | Edilma            | Álvarez Bonilla   | 3236902029
 463842401  | Rocío             | Álvarez Cárdenas  | 3047898372
 788628125  | Mónica            | Álvarez Carrillo  | 3059237130
 939635303  | Marleny           | Álvarez Correa    | 3232410671
 1089620113 | Myriam Yuliana    | Arango Narváez    | 3178267449
 172555225  | Edilma Gabriela   | Arias Álvarez     | 3213658326
 550696749  | Viviana           | Arias Ramírez     | 3204850778
 797017145  | Isabella          | Barrera Cabrera   | 3041744702
 659942505  | Mariana           | Barrera Correa    | 3126485460
 ...
(111 rows)
```

Literal b)

Asociados del municipio 54518 sin correo electrónico registrado. Archivo: `ex_06_b.sql`.

```
SELECT
    T1.document_number AS document,
    T1.first_name AS first_name,
    T1.last_name AS last_name,
    T1.phone_number AS phone,
    T1.email AS email
FROM
    farm.farmers AS T1
WHERE
    T1.municipality_code = '54518'
  AND
    T1.email IS NULL
ORDER BY
    T1.last_name ASC, T1.first_name ASC;
```

Resultado:

```
  document  |    first_name    |     last_name      |   phone    | email
------------+------------------+--------------------+------------+-------
 768461420  | Marleny          | Agudelo Sandoval   | 3242280898 |
 939635303  | Marleny          | Álvarez Correa     | 3232410671 |
 302223305  | Camilo           | Arias Mora         | 3045535545 |
 550696749  | Viviana          | Arias Ramírez      | 3204850778 |
 797017145  | Isabella         | Barrera Cabrera    | 3041744702 |
 483692716  | Ramiro           | Barrera Villamizar | 3056539829 |
 718906996  | Nelson           | Bedoya Vélez       | 3194564518 |
 21358854   | Manuela Yolanda  | Benavides Delgado  | 3229418470 |
 737383707  | Yesenia          | Bermúdez Henao     | 3141005257 |
 1015411133 | Santiago         | Bermúdez Ruiz      | 3003993000 |
 ...
(65 rows)
```

Literal c)

Asociados del municipio 54518 con membresía inactiva, del más antiguo al más reciente. Archivo: `ex_06_c.sql`.

```
SELECT
    T1.document_number AS document,
    T1.first_name AS first_name,
    T1.last_name AS last_name,
    T1.phone_number AS phone,
    T1.membership_date AS membership_date,
    T1.is_active AS is_active
FROM farm.farmers AS T1
WHERE T1.municipality_code = '54518'
  AND T1.is_active = FALSE
ORDER BY T1.membership_date ASC;
```

Resultado:

```
  document  |    first_name    |     last_name      |   phone    | membership_date | is_active
------------+------------------+--------------------+------------+-----------------+-----------
 230971064  | Manuel           | Agudelo Jiménez    | 3219718862 | 1997-03-29      | f
 534023693  | Myriam           | Valencia Moreno    | 3241389451 | 1998-11-03      | f
 33787285   | Rubén            | Ramos Chamorro     | 3244273375 | 1999-03-02      | f
 1135293825 | Abelardo         | Pérez Castro       | 3245612074 | 2003-06-19      | f
 478755763  | Nicolás          | Rivera Duque       | 3029310183 | 2003-08-11      | f
 160538872  | Liliana          | Mosquera Uribe     | 3119365364 | 2007-09-27      | f
 223974138  | Rodrigo Nelson   | Cabrera Villamizar | 3502554241 | 2009-05-02      | f
 10424994   | Cristian         | Herrera Zuluaga    | 3242295228 | 2010-04-28      | f
 1015411133 | Santiago         | Bermúdez Ruiz      | 3003993000 | 2010-11-30      | f
 1090968710 | Ruth             | Rodríguez Pérez    | 3021808053 | 2010-12-11      | f
 ...
(21 rows)
```

Ejercicio 7. BETWEEN y ORDER BY múltiple — Fincas de altura

Tabla: `farm.farms`.

Literal a)

Fincas del municipio 54518 por encima de 1800 m.s.n.m., de mayor a menor altitud. Archivo: `ex_07_a.sql`.

```
SELECT
    T1.farm_name AS farm,
    T1.rural_district AS rural_district,
    T1.total_area_ha AS area_ha,
    T1.altitude_masl AS altitude_masl
FROM farm.farms AS T1
WHERE T1.municipality_code = '54518'
  AND T1.altitude_masl > 1800
ORDER BY T1.altitude_masl DESC;
```

Resultado:

```
      farm      |    rural_district    | area_ha | altitude_masl
----------------+----------------------+---------+---------------
 Santa Clara    | La Legua             |    2.62 |          2300
 El Guayabo     | El Rosal             |    1.21 |          2300
 La Cascada     | Caima                |    2.49 |          2300
 El Vergel      | Sabagua              |    1.11 |          2300
 La Isabela     | García               |    1.36 |          2275
 El Refugio     | Monteadentro         |    4.89 |          2242
 Cafetal Viejo  | Caima                |    5.29 |          2229
 San José       | Alcaparral           |    3.12 |          2203
 El Refugio     | Sabaneta Baja        |    1.93 |          2195
 Villa Hermosa  | Peñas                |    0.30 |          2181
 ...
(108 rows)
```

Literal b)

Fincas entre 1600 y 1800 m.s.n.m., por vereda ascendente y área descendente. Archivo: `ex_07_b.sql`.

```
SELECT
    T1.farm_name AS farm,
    T1.rural_district AS rural_district,
    T1.total_area_ha AS area_ha,
    T1.altitude_masl AS altitude_masl
FROM farm.farms AS T1
WHERE T1.municipality_code = '54518'
  AND T1.altitude_masl BETWEEN 1600 AND 1800
ORDER BY T1.rural_district ASC, T1.total_area_ha DESC;
```

Resultado:

```
       farm       |    rural_district    | area_ha | altitude_masl
------------------+----------------------+---------+---------------
 Los Pinos        | Alcaparral           |    1.67 |          1771
 Miraflores       | Alcaparral           |    1.29 |          1636
 Monteverde       | Alto Grande          |    2.59 |          1794
 La Loma          | Alto Grande          |    2.09 |          1667
 Bellavista       | Alto Grande          |    1.80 |          1704
 El Refugio       | Alto Grande          |    1.23 |          1661
 El Jardín        | Cabecera Municipal   |    3.26 |          1620
 El Edén          | Cabecera Municipal   |    2.89 |          1603
 La Milagrosa     | Cabecera Municipal   |    2.65 |          1643
 La Divisa        | Cabecera Municipal   |    1.32 |          1765
 ...
(104 rows)
```

Literal c)

Fincas del municipio 54518 con latitud o longitud nulas. Archivo: `ex_07_c.sql`.

```
SELECT
    T1.farm_name AS farm,
    T1.rural_district AS rural_district,
    T1.latitude AS latitude,
    T1.longitude AS longitude
FROM farm.farms AS T1
WHERE T1.municipality_code = '54518'
  AND (T1.latitude IS NULL OR T1.longitude IS NULL);
```

Resultado:

```
       farm       |    rural_district    | latitude | longitude
------------------+----------------------+----------+-----------
 El Recuerdo      | Alto Grande          |          |
 La Esperanza     | El Palchal           |          |
 Las Delicias     | Caima                |          |
 Los Arrayanes    | Cabecera Municipal   |          |
 El Brillante     | Peñas                |          |
 El Porvenir      | Fontibón             |          |
 La Florida       | Peñas                |          |
 Los Arrayanes    | Navarro              |          |
 El Guayabo       | San Francisco        |          |
 El Encanto       | Fontibón             |          |
 ...
(153 rows)
```

Ejercicio 8. IN y columnas calculadas — Entregas del día

Tabla: `buy.deliveries`.

Literal a)

Entregas del 30 de septiembre de 2026, ordenadas por hora de recepción. Archivo: `ex_08_a.sql`.

```
SELECT
    T1.delivery_id AS delivery,
    T1.farm_id AS farm_id,
    T1.collection_point_id AS collection_point_id,
    T1.delivered_at AS received_at,
    T1.coffee_type AS coffee_type,
    T1.gross_weight_kg AS gross_weight_kg
FROM buy.deliveries AS T1
WHERE T1.delivery_id LIKE 'DLV-20260930-%'
ORDER BY T1.delivered_at ASC;
```

Resultado:

```
      delivery       | farm_id | collection_point_id |     received_at     |  coffee_type  | gross_weight_kg
---------------------+---------+---------------------+---------------------+---------------+-----------------
 DLV-20260930-000001 |  121925 |                  68 | 2026-09-30 06:01:10 | dry_parchment |           51.53
 DLV-20260930-000002 |   39639 |                  48 | 2026-09-30 06:03:32 | dry_parchment |          124.98
 DLV-20260930-000003 |  228920 |                  80 | 2026-09-30 06:04:40 | wet_parchment |          112.33
 DLV-20260930-000004 |    2649 |                  90 | 2026-09-30 06:07:26 | dry_parchment |          138.29
 DLV-20260930-000005 |  163723 |                  10 | 2026-09-30 06:07:51 | dry_parchment |          256.42
 DLV-20260930-000006 |   58408 |                 170 | 2026-09-30 06:11:03 | dry_parchment |          139.81
 DLV-20260930-000007 |  201601 |                  57 | 2026-09-30 06:15:25 | dry_parchment |          404.97
 DLV-20260930-000008 |  135490 |                  53 | 2026-09-30 06:16:53 | dry_parchment |          175.92
 DLV-20260930-000009 |  108211 |                  54 | 2026-09-30 06:17:34 | dry_parchment |           56.80
 DLV-20260930-000010 |   37288 |                  67 | 2026-09-30 06:20:30 | dry_parchment |          269.43
 ...
(300 rows)
```

Literal b)

Entregas rechazadas de ese día, con el peso promedio por bulto. Archivo: `ex_08_b.sql`.

```
SELECT
    T1.delivery_id AS delivery,
    T1.farm_id AS farm_id,
    T1.collection_point_id AS collection_point_id,
    T1.delivered_at AS received_at,
    T1.coffee_type AS coffee_type,
    T1.gross_weight_kg AS gross_weight_kg,
    T1.bag_count AS bag_count,
    T1.gross_weight_kg / T1.bag_count AS avg_kg_per_bag
FROM buy.deliveries AS T1
WHERE T1.delivery_id LIKE 'DLV-20260930-%'
  AND T1.status = 'rejected'
ORDER BY T1.delivered_at ASC;
```

Resultado:

```
      delivery       | farm_id | collection_point_id |     received_at     |  coffee_type  | gross_weight_kg | bag_count |   avg_kg_per_bag
---------------------+---------+---------------------+---------------------+---------------+-----------------+-----------+--------------------
 DLV-20260930-000006 |   58408 |                 170 | 2026-09-30 06:11:03 | dry_parchment |          139.81 |         3 | 46.6033333333333333
 DLV-20260930-000070 |   69157 |                   3 | 2026-09-30 08:19:57 | dry_parchment |          147.13 |         3 | 49.0433333333333333
 DLV-20260930-000083 |   66435 |                   2 | 2026-09-30 09:20:30 | dry_parchment |           89.95 |         2 | 44.9750000000000000
 DLV-20260930-000109 |  216626 |                   6 | 2026-09-30 10:04:15 | dry_parchment |          311.88 |         5 | 62.3760000000000000
 DLV-20260930-000147 |   77310 |                 103 | 2026-09-30 11:17:45 | wet_parchment |           66.42 |         1 | 66.4200000000000000
 DLV-20260930-000165 |  107038 |                 162 | 2026-09-30 12:15:01 | dry_parchment |          348.59 |         6 | 58.0983333333333333
 DLV-20260930-000214 |  223116 |                  44 | 2026-09-30 13:53:56 | dry_parchment |          177.63 |         3 | 59.2100000000000000
 DLV-20260930-000223 |  216415 |                  45 | 2026-09-30 14:19:21 | dry_parchment |          338.32 |         7 | 48.3314285714285714
 DLV-20260930-000253 |  143807 |                 102 | 2026-09-30 15:05:34 | dry_parchment |          253.15 |         5 | 50.6300000000000000
(9 rows)
```

Literal c)

Entregas de ese día de tipo wet_parchment o cherry, por peso bruto descendente. Archivo: `ex_08_c.sql`.

```
SELECT
    T1.delivery_id AS delivery,
    T1.farm_id AS farm_id,
    T1.collection_point_id AS collection_point_id,
    T1.delivered_at AS received_at,
    T1.coffee_type AS coffee_type,
    T1.gross_weight_kg AS gross_weight_kg
FROM buy.deliveries AS T1
WHERE T1.delivery_id LIKE 'DLV-20260930-%'
  AND T1.coffee_type IN ('wet_parchment', 'cherry')
ORDER BY T1.gross_weight_kg DESC;
```

Resultado:

```
      delivery       | farm_id | collection_point_id |     received_at     |  coffee_type  | gross_weight_kg
---------------------+---------+---------------------+---------------------+---------------+-----------------
 DLV-20260930-000081 |   19338 |                 117 | 2026-09-30 09:17:37 | cherry        |          475.13
 DLV-20260930-000187 |  231839 |                 159 | 2026-09-30 12:47:52 | cherry        |          339.30
 DLV-20260930-000105 |   13478 |                  37 | 2026-09-30 09:57:09 | wet_parchment |          277.63
 DLV-20260930-000028 |  182397 |                 133 | 2026-09-30 07:13:46 | wet_parchment |          258.00
 DLV-20260930-000133 |   44726 |                  59 | 2026-09-30 10:51:03 | cherry        |          254.83
 DLV-20260930-000094 |   99627 |                 133 | 2026-09-30 09:31:20 | cherry        |          246.96
 DLV-20260930-000129 |   89312 |                  76 | 2026-09-30 10:44:39 | wet_parchment |          206.71
 DLV-20260930-000161 |  101318 |                  24 | 2026-09-30 12:06:51 | wet_parchment |          197.04
 DLV-20260930-000046 |   85553 |                 140 | 2026-09-30 07:30:47 | wet_parchment |          186.73
 DLV-20260930-000248 |  215660 |                 166 | 2026-09-30 14:59:13 | cherry        |          159.89
 ...
(22 rows)
```

Ejercicio 9. Rangos de calidad — Análisis de laboratorio

Tabla: `buy.quality_analyses`.

Literal a)

Análisis del 30 de septiembre de 2026 con cup_score mayor o igual a 85, de mayor a menor. Archivo: `ex_09_a.sql`.

```
SELECT
    T1.analysis_id AS analysis,
    T1.delivery_id AS delivery,
    T1.moisture_pct AS moisture_pct,
    T1.yield_factor AS yield_factor,
    T1.cup_score AS cup_score
FROM buy.quality_analyses AS T1
WHERE T1.analyzed_at >= '2026-09-30'
  AND T1.analyzed_at < '2026-10-01'
  AND T1.cup_score >= 85
ORDER BY T1.cup_score DESC;
```

Resultado:

```
      analysis      |      delivery       | moisture_pct | yield_factor | cup_score
--------------------+---------------------+--------------+--------------+-----------
 QA-20260930-000102 | DLV-20260930-000155 |        12.31 |        94.01 |     87.28
 QA-20260930-000082 | DLV-20260930-000111 |        12.38 |        94.29 |     87.18
 QA-20260930-000137 | DLV-20260930-000295 |        12.35 |        91.07 |     87.14
 QA-20260930-000116 | DLV-20260930-000208 |        11.07 |        93.78 |     87.07
 QA-20260930-000118 | DLV-20260930-000256 |        11.16 |        92.54 |     87.04
 QA-20260930-000030 | DLV-20260930-000089 |        10.70 |        91.53 |     86.31
 QA-20260930-000089 | DLV-20260930-000203 |        11.84 |        91.41 |     86.01
 QA-20260930-000028 | DLV-20260930-000064 |        11.48 |        94.23 |     85.62
 QA-20260930-000110 | DLV-20260930-000182 |        10.64 |        88.18 |     85.25
 QA-20260930-000079 | DLV-20260930-000085 |        11.55 |        93.28 |     85.15
 ...
(11 rows)
```

Literal b)

Análisis de ese día con humedad fuera del rango 10%–12%, con analista y observaciones. Archivo: `ex_09_b.sql`.

```
SELECT
    T1.analysis_id AS analysis,
    T1.delivery_id AS delivery,
    T1.moisture_pct AS moisture_pct,
    T1.analyst_name AS analyst_name,
    T1.observations AS observations
FROM buy.quality_analyses AS T1
WHERE T1.analyzed_at >= '2026-09-30'
  AND T1.analyzed_at < '2026-10-01'
  AND T1.moisture_pct NOT BETWEEN 10 AND 12;
```

Resultado:

```
      analysis      |      delivery       | moisture_pct |    analyst_name    |                          observations
--------------------+---------------------+--------------+--------------------+-----------------------------------------------------------
 QA-20260930-000026 | DLV-20260930-000005 |        12.19 | Wilson Galvis      |
 QA-20260930-000018 | DLV-20260930-000006 |        17.24 | Nicolás Serrano    | Café con humedad alta, requiere más secado.
 QA-20260930-000017 | DLV-20260930-000012 |        12.15 | Natalia Trujillo   |
 QA-20260930-000033 | DLV-20260930-000027 |        12.41 | Sara Sánchez       |
 QA-20260930-000013 | DLV-20260930-000038 |        12.28 | Adriana Losada     |
 QA-20260930-000005 | DLV-20260930-000040 |        12.22 | Iván Castro        |
 QA-20260930-000027 | DLV-20260930-000043 |        12.07 | Germán González    | Caficultor solicita contramuestra.
 QA-20260930-000023 | DLV-20260930-000046 |        12.23 | Alejandra Torres   | Café de buena presentación, sin olores extraños.
 QA-20260930-000012 | DLV-20260930-000057 |        12.16 | Guillermo Acevedo  |
 QA-20260930-000022 | DLV-20260930-000070 |        12.95 | Santiago Rojas     | Café con humedad alta, requiere más secado.
 ...
(40 rows)
```

Literal c)

Análisis no aceptados de ese día con observaciones, por porcentaje de defectos descendente. Archivo: `ex_09_c.sql`.

```
SELECT
    T1.analysis_id AS analysis,
    T1.delivery_id AS delivery,
    T1.is_accepted AS is_accepted,
    T1.observations AS observations,
    T1.defects_pct AS defects_pct
FROM buy.quality_analyses AS T1
WHERE T1.analyzed_at >= '2026-09-30'
  AND T1.analyzed_at < '2026-10-01'
  AND T1.is_accepted = FALSE
  AND T1.observations IS NOT NULL
ORDER BY T1.defects_pct DESC;
```

Resultado:

```
      analysis      |      delivery       | is_accepted |               observations                 | defects_pct
--------------------+---------------------+-------------+--------------------------------------------+-------------
 QA-20260930-000104 | DLV-20260930-000214 | f           | Olor a fermento en la muestra.             |       19.54
 QA-20260930-000068 | DLV-20260930-000165 | f           | Muestra tomada por duplicado.              |       17.29
 QA-20260930-000119 | DLV-20260930-000253 | f           | Exceso de impurezas y pasilla.             |       15.59
 QA-20260930-000086 | DLV-20260930-000147 | f           | Olor a fermento en la muestra.             |       13.21
 QA-20260930-000113 | DLV-20260930-000223 | f           | Café sobresecado, grano quebradizo.        |       12.94
 QA-20260930-000022 | DLV-20260930-000070 | f           | Café con humedad alta, requiere más secado. |        9.75
 QA-20260930-000018 | DLV-20260930-000006 | f           | Café con humedad alta, requiere más secado. |        7.88
(7 rows)
```

Ejercicio 10. Columnas generadas — Liquidaciones pendientes

Tabla: `pay.settlements`.

Literal a)

Liquidaciones del 30 de septiembre de 2026 con peso, precio, valor bruto, deducciones y valor neto. Archivo: `ex_10_a.sql`.

```
SELECT
    T1.settlement_id AS settlement,
    T1.delivery_id AS delivery,
    T1.net_weight_kg AS net_weight_kg,
    T1.price_per_kg_applied AS price_per_kg,
    T1.gross_amount AS gross_cop,
    T1.deductions_amount AS deductions_cop,
    T1.net_amount AS net_cop
FROM pay.settlements AS T1
WHERE T1.settled_at >= '2026-09-30'
  AND T1.settled_at < '2026-10-01';
```

Resultado:

```
     settlement      |      delivery       | net_weight_kg | price_per_kg | gross_cop  | deductions_cop |  net_cop
---------------------+---------------------+---------------+--------------+------------+----------------+------------
 STL-20260930-000025 | DLV-20260927-000082 |        180.06 |     22983.00 | 4138318.98 |      182938.77 | 3955380.21
 STL-20260930-000011 | DLV-20260927-000088 |        211.64 |     22841.00 | 4834069.24 |           0.00 | 4834069.24
 STL-20260930-000018 | DLV-20260927-000097 |        216.32 |     21645.00 | 4682246.40 |           0.00 | 4682246.40
 STL-20260930-000002 | DLV-20260927-000104 |        163.44 |     22367.00 | 3655662.48 |      201901.82 | 3453760.66
 STL-20260930-000035 | DLV-20260927-000128 |        237.30 |     22840.00 | 5419932.00 |           0.00 | 5419932.00
 STL-20260930-000027 | DLV-20260927-000144 |        139.52 |     21607.00 | 3014608.64 |           0.00 | 3014608.64
 STL-20260930-000003 | DLV-20260927-000169 |        338.42 |     20409.00 | 6906813.78 |      344352.08 | 6562461.70
 STL-20260930-000007 | DLV-20260927-000177 |        226.45 |     20916.00 | 4736428.20 |      167394.38 | 4569033.82
 STL-20260930-000030 | DLV-20260927-000197 |         39.43 |     23273.00 |  917654.39 |           0.00 |  917654.39
 STL-20260930-000005 | DLV-20260927-000198 |        158.94 |     21608.00 | 3434375.52 |           0.00 | 3434375.52
 ...
(69 rows)
```

Literal b)

Liquidaciones de ese día en estado pending, por valor neto descendente. Archivo: `ex_10_b.sql`.

```
SELECT
    T1.settlement_id AS settlement,
    T1.delivery_id AS delivery,
    T1.net_weight_kg AS net_weight_kg,
    T1.gross_amount AS gross_cop,
    T1.deductions_amount AS deductions_cop,
    T1.net_amount AS net_cop,
    T1.status AS status
FROM pay.settlements AS T1
WHERE T1.settled_at >= '2026-09-30'
  AND T1.settled_at < '2026-10-01'
  AND T1.status = 'pending'
ORDER BY T1.net_amount DESC;
```

Resultado:

```
     settlement      |      delivery       | net_weight_kg | gross_cop  | deductions_cop |  net_cop   | status
---------------------+---------------------+---------------+------------+----------------+------------+---------
 STL-20260930-000062 | DLV-20260929-000235 |        283.06 | 6171557.18 |           0.00 | 6171557.18 | pending
 STL-20260930-000013 | DLV-20260928-000004 |        236.95 | 5293226.05 |      205370.50 | 5087855.55 | pending
 STL-20260930-000069 | DLV-20260930-000204 |        231.21 | 5246617.32 |      232910.66 | 5013706.66 | pending
 STL-20260930-000021 | DLV-20260928-000105 |        219.83 | 4912541.01 |           0.00 | 4912541.01 | pending
 STL-20260930-000011 | DLV-20260927-000088 |        211.64 | 4834069.24 |           0.00 | 4834069.24 | pending
 STL-20260930-000018 | DLV-20260927-000097 |        216.32 | 4682246.40 |           0.00 | 4682246.40 | pending
 STL-20260930-000007 | DLV-20260927-000177 |        226.45 | 4736428.20 |      167394.38 | 4569033.82 | pending
 STL-20260930-000068 | DLV-20260928-000311 |        192.15 | 4341052.80 |           0.00 | 4341052.80 | pending
 STL-20260930-000033 | DLV-20260929-000007 |        183.79 | 4073521.56 |           0.00 | 4073521.56 | pending
 STL-20260930-000016 | DLV-20260928-000207 |        180.13 | 3923051.27 |           0.00 | 3923051.27 | pending
 ...
(28 rows)
```

Literal c)

Liquidaciones de ese día sin deducciones, con la columna calculada check_gross_cop. Archivo: `ex_10_c.sql`.

```
SELECT
    T1.settlement_id AS settlement,
    T1.delivery_id AS delivery,
    T1.net_weight_kg AS net_weight_kg,
    T1.price_per_kg_applied AS price_per_kg,
    T1.deductions_amount AS deductions_cop,
    T1.gross_amount AS gross_cop,
    T1.net_weight_kg * T1.price_per_kg_applied AS check_gross_cop
FROM pay.settlements AS T1
WHERE T1.settled_at >= '2026-09-30'
  AND T1.settled_at < '2026-10-01'
  AND T1.deductions_amount = 0;
```

Resultado:

```
     settlement      |      delivery       | net_weight_kg | price_per_kg | deductions_cop | gross_cop  | check_gross_cop
---------------------+---------------------+---------------+--------------+----------------+------------+-----------------
 STL-20260930-000016 | DLV-20260928-000207 |        180.13 |     21779.00 |           0.00 | 3923051.27 |    3923051.2700
 STL-20260930-000038 | DLV-20260928-000241 |        247.93 |     21622.00 |           0.00 | 5360742.46 |    5360742.4600
 STL-20260930-000068 | DLV-20260928-000311 |        192.15 |     22592.00 |           0.00 | 4341052.80 |    4341052.8000
 STL-20260930-000011 | DLV-20260927-000088 |        211.64 |     22841.00 |           0.00 | 4834069.24 |    4834069.2400
 STL-20260930-000018 | DLV-20260927-000097 |        216.32 |     21645.00 |           0.00 | 4682246.40 |    4682246.4000
 STL-20260930-000035 | DLV-20260927-000128 |        237.30 |     22840.00 |           0.00 | 5419932.00 |    5419932.0000
 STL-20260930-000027 | DLV-20260927-000144 |        139.52 |     21607.00 |           0.00 | 3014608.64 |    3014608.6400
 STL-20260930-000030 | DLV-20260927-000197 |         39.43 |     23273.00 |           0.00 |  917654.39 |     917654.3900
 STL-20260930-000005 | DLV-20260927-000198 |        158.94 |     21608.00 |           0.00 | 3434375.52 |    3434375.5200
 STL-20260930-000042 | DLV-20260928-000048 |        300.18 |     23455.00 |           0.00 | 7040721.90 |    7040721.9000
 ...
(37 rows)
```

Explicación:

Las dos columnas coinciden porque en el fondo son la misma cuenta: peso neto por precio aplicado por kilo. La diferencia es quién la hace. `gross_amount` es una columna GENERATED, o sea que PostgreSQL la calcula sola y la guarda cuando se registra la liquidación, mientras que `check_gross_cop` la calculé yo en el SELECT con esas mismas dos columnas. Por eso tiene que dar lo mismo. Lo único que se ve distinto es que `check_gross_cop` sale con 4 decimales y `gross_amount` con 2, pero al comparar las 37 filas el valor es igual y solo cambian los ceros del final.